

Plan: Overworked Americans take-home

What I am building, in plain English, for whoever wants to follow

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The whole story in one paragraph. Q1 says: in the data, EU hours fell vs. US, and EU tax wedges rose vs. US – but the cross-sectional fit between hours and taxes is too noisy ($R^2 = 0.16$) for taxes to be the whole story. Q2 puts the simplest labour-supply model (Prescott log-log) on this data and asks: what elasticity would the model need to fit the EU-US gap? The answer is ~ 0.7 – much larger than what the micro labour-supply literature gives us (~ 0.3 , Chetty–Guren–Manoli–Weber 2011). So either the macro elasticity really is that big (the original Prescott bet), or there are *other things* – institutional or cultural – pulling EU hours down at common τ . We pursue the second route. Q4 asks: what if Europe has hours regulation that the US lacks, and that regulation is binding? Q5 asks: what if Europe has unions and the US essentially does not, and unions bargain hours toward the level workers actually prefer? Q6 asks: what if leisure in Europe is more valuable because everyone else is also on holiday? Q7 puts all the channels in one frame and asks how much each one contributes to the US–EU welfare gap.

1. The setup, in plain English

We will use the same firm–worker pair throughout Q4–Q5–Q6, with one small twist per question. The pair is described by two simple payoff functions:

$$\underbrace{V_w(h) = y \cdot h - \frac{\alpha}{2} h^2}_{\text{worker payoff}}, \quad \underbrace{V_f(h) = A \cdot h - \frac{\beta}{2} h^2 - w \cdot h}_{\text{firm payoff}}.$$

The worker’s payoff is income minus a quadratic disutility of work. The income piece is $y \cdot h$ where $y \equiv (1 - \tau)w$ is take-home hourly wage and h is hours – so $y \cdot h$ is the worker surplus function the question prompt asks for, with the tax wedge built in. The firm’s payoff is revenue (concave in h , with diminishing returns) minus the wage bill.

Three reference values of h matter throughout the rest of the problem:

- $h^* \equiv \arg \max V_w = \frac{(1 - \tau)w}{\alpha}$ – the hours *the worker would choose* if free to pick.
- $h^{\text{firm}} \equiv \arg \max V_f = \frac{A - w}{\beta}$ – the hours *the firm would choose* given a fixed wage.
- $h^{\text{soc}} \equiv \arg \max [V_w + V_f] = \frac{(1 - \tau)w + (A - w)}{\alpha + \beta}$ – the hours that maximize *joint surplus* (the social planner’s choice).

At the baseline calibration ($\alpha = \beta = 2$, $A = 2$, $w = 1$, $\tau = 0.4$) these are $h^* = 0.30$, $h^{\text{soc}} = 0.40$, $h^{\text{firm}} = 0.50$. The whole question is about *which of these three the equilibrium settles at*, and how to move it.

1.1 Why we need an institutional/cultural channel at all

Imagine US and EU face the same wage w and the same tax τ . Two clean benchmarks:

- Under *perfect Walrasian clearing*, the worker freely picks $h^*(\tau)$ in both countries. There is no US–EU hours gap at common τ .
- Under *firm-power* (the firm sets h unilaterally, the worker accepts), the firm picks h^{firm} in both countries. Again no US–EU gap at common τ .

Neither pure benchmark explains why EU works fewer hours than the US at the same τ . So we need *something institutional or cultural* that breaks the symmetry. That “something” is what Q4, Q5, Q6 explore – one candidate per question.

1.2 What we deliberately leave out (footnotes-worth)

We do not model the *extensive margin* – whether people work at all – because participation depends on joint-vs-individual income tax filing, public childcare, employment-protection legislation, and the welfare system, all of which is a separate body of theory. The prompt asks specifically about *hours worked*, so we focus on the intensive margin (Bick–Brüggemann–Fuchs–Schündeln 2019 anyway show that the extensive margin and intensive margin contribute roughly equally to the EU–US H/N gap, but each has its own machinery).

We also do not model *contract stickiness* – the fact that in many European countries, weekly hours per worker are set in a sectoral collective contract that is not renegotiated for years even when preferences or productivity change. This is a real friction documented in Cahuc–Carcillo (2014, IZA WP) and the EU Labour Force Survey, but it is observationally equivalent (in our static framework) to a binding hours cap, and is therefore captured implicitly through the cap mechanism of Q4.

We also do not model *sectoral composition*. Larger public sectors in Europe imply more workers on contracts that are sticky for political-economy reasons unrelated to firm-profit maximisation; public-sector hours decisions are governed by a different mechanism (collective bargaining over service provision). Our results should be read as applying to the private-sector portion of the labour market.

2. Q1 – the data inconsistency

We open with three panels of data: (a) hours per worker, percent deviation from US, 1970–2024; (b) effective labour wedge τ_h , percent deviation from US, 1970–2007 (McDaniel 2011); (c) composite tax wedge τ vs. hours per worker, 2023 cross-section across 19 OECD countries.

Panels (a) and (b) tell a coherent story: hours fell against the US after 1970, wedges rose against the US, and the timing matches roughly. Panel (c) is where things get interesting. The OLS slope is signed correctly (−9.9 hours per percentage point of wedge), but the $R^2 = 0.16$ and the residual standard deviation is ~ 158 hours – about one-third of the entire DE–US hours gap. Italy and Greece have lower wedges than France yet work *more* hours per worker; Belgium, Germany, Austria, France span ~ 200 hours within a 6 pp wedge band.

The cross-section of taxes vs. hours is too noisy for taxes to be the whole story. Either the within-country time-series response of hours to taxes is much larger than the cross-section delivers (the *Prescott–CGM puzzle* we go after in Q2), or the cross-section is corrupted by country-specific institutional or cultural frictions that the tax wedge does not capture (the *Q4–Q6 candidates*).

3. Q2 – the Prescott model and the elasticity puzzle

The simplest model of labour supply under taxation is Prescott’s (2004): a representative household with log–log preferences, $u(c, h) = \log c + \alpha \log(1 - h)$, faces a wage w , a labour tax, a consumption tax, and a balanced lump-sum rebate. The first thing the model gives us is a clean composite tax wedge $1 - \tau \equiv (1 - \tau_l)/(1 + \tau_c)$ that absorbs both the labour and the consumption

tax (since workers supply labour only to consume, the consumption tax is isomorphic to a labour tax). The first-order condition then gives the worker's preferred hours as a closed-form function of τ :

$$h^*(\tau) = \frac{1 - \tau}{\alpha + 1 - \tau}, \quad \varepsilon_{h,1-\tau}^M = \frac{\alpha}{\alpha + 1 - \tau}.$$

3.1 The calibrated model fits, but at the wrong elasticity

Anchor $h^{US} = 0.33$ at $\tau^{US} = 0.35$. Inverting the formula gives $\alpha = 1.32$. With this single parameter pinned down, the model's prediction at $\tau^{EU} = 0.56$ is $h^{EU} = 0.44/1.76 = 0.250$ – and the data point is 0.247. *The model essentially fits the cross-section.*

So why is there a puzzle? Because the same calibration pins down the labour-supply elasticity at ~ 0.67 – 0.75 along the supply curve, and the micro labour-supply literature (Chetty–Guren–Manoli–Weber 2011, summarising the quasi-experimental evidence) reports a Hicksian intensive elasticity of about 0.30 for prime-age workers. *The calibrated α implies an elasticity that is roughly $2.5\times$ the micro estimate.*

The puzzle is therefore not that the model under-explains the data, but that the calibrated parameter is inconsistent with quasi-experimental evidence on individual labour supply. Either:

- the macro elasticity really *is* that large, and the micro literature is missing something (life-cycle aggregation, Rogerson–Wallenius 2009; tax progressivity, HSV 2017; participation elasticity, BFL 2018 – all genuine modern channels that close part of the gap);
- or there are *other channels* that produce the cross-country hours gap at low individual elasticity, so the macro α is absorbing things that should be modelled separately as institutions or culture.

3.2 The decomposition that motivates Q4–Q7

Pin α at the micro-consistent value $\alpha^{\text{micro}} = 0.279$ (the value that delivers $\varepsilon^M = 0.30$ at the US point). Now the Prescott model *alone* predicts US hours of 0.700 and EU hours of 0.612 – both much higher than observed, in absolute terms, but the cross-country log gap is only -13.4% . The data shows a -29.0% log gap. So at the micro elasticity, taxes alone explain about 46% of the cross-country hours gap; the remaining 54% – a 15.6 percentage-point log residual – must come from somewhere outside the Prescott mechanism.

This is the load-bearing observation. The institutional and cultural channels of Q4–Q6 are not just nice-to-have additions to a story that Prescott already tells; they are quantitatively necessary at any plausible labour-supply elasticity. *This is what motivates Q4, Q5, Q6 as serious candidates rather than rhetorical ones.*

4. Q4 – hours regulation as a quantity instrument

The first candidate is straightforward: *governments in Europe constrain hours, while the US essentially does not.* The 35-hour week in France, the EU Working Time Directive, sectoral hour caps in Germany and Italy – all are quantity instruments that, if binding, keep equilibrium hours below what they would otherwise be.

4.1 The model: a cap on top of the firm-power baseline

We use the firm-power baseline as our reference point: in the absence of regulation, the firm picks $h^{\text{firm}} = (A - w)/\beta = 0.5$ and the worker accepts (think of this as the result of a long-term contract that is sticky in the EU; in the US the same outcome arises from a thinner labour market). The government imposes a cap $\bar{h}$. The constrained equilibrium is

$$h^{eq}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}.$$

If the cap is above h^{firm} it is non-binding; if below, equilibrium hours equal the cap.

4.2 The intuition: it's an externality problem

The welfare condition for the cap is exactly the Pigouvian-tax analogue from the externalities chapter. Joint surplus $W(h) = V_w + V_f$ is concave with peak at $h^{\text{soc}} = 0.40$. So moving from $h^{\text{firm}} = 0.50$ down toward $h^{\text{soc}} = 0.40$ *increases* joint welfare – the firm is producing too many hours relative to the joint optimum, and pulling down a little corrects the distortion. But moving *past* h^{soc} – e.g., to a cap at $h = 0.25$ – now overshoots in the other direction: too few hours, joint surplus falls below where it was at the firm-power baseline.

The clean result: the cap is welfare-improving iff $\bar{h} \in (2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}})$, which at our $\alpha = \beta$ baseline simplifies to

$$\bar{h} \in (h^*, h^{\text{firm}}) \text{ for the cap to be welfare-improving}$$

Equivalently: any cap that pulls hours down toward h^{soc} helps; any cap that pushes hours below the worker's individual optimum h^* hurts (the worker is now being forced to work *less* than they themselves would prefer at this wage).

4.3 What this says about US vs. EU

In our framework, the question is whether the EU cap binds and the US cap does not. The Aubry 35-hour reform in France maps to roughly $\bar{h} \approx 0.875 \cdot h^{\text{firm}} = 0.4375$ – comfortably inside the welfare-improving band. The empirical evidence (Estevão-Sá 2008; Chemin-Wasmer 2009) confirms a $\sim 4\%$ hours decline in covered French sectors with near-zero employment effect: a cap that bound, that worked, and that did not overshoot.

The US has the FLSA 40-hour overtime threshold but it rarely binds at the median margin – so for our purposes the US is at h^{firm} and the EU is at $\bar{h}$. This is one source of the EU–US hours gap at common τ .

4.4 Why the cap can be the wrong tool

There is a real cost to the cap that the worked example glosses over: *workers are heterogeneous in α* . A single national cap helps high- α (leisure-loving) workers, whose individual h^* is far below h^{firm} anyway, and hurts low- α (income-loving) workers, whose individual h^* may be above $\bar{h}$. The cap therefore redistributes from low- α workers to high- α workers, which may or may not be what a planner wants. We flag this in the body but do not model it explicitly.

5. Q5 – a single union (the heart of the modelling story)

The second candidate is more elegant. Caps are blunt: a government bureaucrat picks a single number and applies it to everyone, with no guarantee that it sits in the welfare-improving band. *What if the constraint on hours could emerge endogenously from the interaction between workers and firms?* That is what unions do.

5.1 The setup: one union, all workers inside, dues share s

We model the simplest possible union: a single industry-wide union, all workers are inside, each pays a fixed share $s \in (0, 1)$ of net labour income to the union. The union's payoff is $s \cdot V_w(h)$, so

$$\arg \max_h [s \cdot V_w(h)] = \arg \max_h V_w(h) = h^*.$$

The dues share scales the union's objective but does not move what the union wants. *This is the central modelling point:* the union and the workers are perfectly aligned, no agency problem.

(Compare a union that maximises dues revenue $s \cdot wh$; that union would prefer h^{firm} – dues revenue is increasing in hours – and pull in the wrong direction.)

5.2 Why this is conceptually better than a cap

The cap of Q4 is set by a government planner who must guess h^{soc} correctly. If the planner overshoots, welfare falls. The union of Q5 *endogenously* pulls hours toward h^* , which is inside the welfare-improving band by construction (since h^* is *at* the worker-side boundary of that band). This is the same intuition that makes *Coase* better than *Pigou* when the parties can negotiate: let the bargaining endogenously select the right level, rather than have the government impose a quantity *ex ante*.

The union therefore has the potential to fix the firm-power distortion without requiring the planner to know h^{soc} .

5.3 The variable that does the work: bargaining strength η

The actual hours outcome depends on *how strongly* the union can bargain. Let $\eta \in [0, 1]$ be the worker's bargaining weight in the firm–union negotiation. At $\eta = 0$ (no union influence) the firm picks h^{firm} . At $\eta = 1$ (full union dominance) the union pulls hours all the way to h^* . For intermediate η , the bargained outcome interpolates:

$$h(\eta) = \frac{(1 - \eta)(A - w) + \eta(1 - \tau)w}{(1 - \eta)\beta + \eta\alpha}.$$

At our symmetric baseline ($\alpha = \beta = 2$) this is linear in η .

5.4 The optimality condition for the union to be welfare-improving

Joint surplus $W(\eta)$ is concave in η with peak at the η^{soc} where $h(\eta^{\text{soc}}) = h^{\text{soc}}$. At our baseline this is $\eta^{\text{soc}} = 1/2$ – so the joint optimum is hit when the union has *half* the bargaining weight. The welfare-improving region is

$$\eta \in (0, 2\eta^{\text{soc}}) \text{ for the union bargain to be welfare-improving}$$

which at baseline simplifies to $\eta \in (0, 1)$. *A moderately strong union helps; a maximally strong union pulls hours all the way to h^* , which is below h^{soc} , and total surplus falls back to the no-union level.*

This is the core finding for Q5: **the welfare effect of a union is non-monotone in its strength**. A toothless union does nothing; an all-powerful union over-corrects; a moderate union sits at the joint optimum. The story closely mirrors Q4: over-shoot and under-shoot in either direction destroy welfare.

5.5 What this says about US vs. EU

In the US, bargaining-coverage rates are around 10%. In Italy they are around 80%, in France $\sim 95\%$, Sweden $\sim 90\%$. The order-of-magnitude difference in coverage maps to a several-fold difference in η . So the model predicts:

- **US:** low η , equilibrium hours close to h^{firm} (no union channel doing work);
- **EU:** moderate η , equilibrium hours pulled toward h^* , ideally near h^{soc} where the welfare gain is maximised.

The US–EU hours gap at common τ is then a difference in η , not in α or τ .

5.6 The honest caveat: what happens under canonical right-to-manage

The above is the cleanest pedagogical version. The canonical labour economics literature (Nickell 1982; Layard–Nickell–Jackman 1991) uses *right-to-manage* (RTM): the union and firm bargain over the wage w , and the firm then picks h on its labour-demand curve. Under RTM, the union still pulls outcomes toward the worker side, but it does so by raising w , which has a side effect: the tax wedge τwh rises, and that wedge is dead-weight (lost from $V_w + V_f$). Once you account for this, the canonical result is that W falls monotonically in η – there is no welfare-improving region.

The two readings (sticky- w Coasian intuition vs. flexible- w canonical RTM) give different welfare conclusions because they make different assumptions about what is bargained. The sticky- w reading is the cleanest pedagogical version and probably the closest to the “Coase beats Pigou” intuition the prompt is fishing for. The canonical RTM reading is what the cross-country macro-labour literature actually uses. We present both in the body of the take-home and let the reader decide.

6. Q6 – culture as a leisure bonus (enjoy life more = higher welfare)

The third candidate is the most direct: *Europeans enjoy leisure more than Americans do*. This shows up empirically – subjective wellbeing surveys (Stevenson–Wolfers 2008, BPEA Spring), time-use studies, and revealed preferences in cross-country labour-supply data all point to a higher marginal value of leisure on the European side.

6.1 Two ways to write “culture means people enjoy life more”

There is a real modelling choice here that matters for welfare. Two options:

(i) **Higher disutility of work.** Increase α for the EU worker, so working hurts them more. This delivers h^* down – but also delivers V_w down at every h . This is the “Europeans suffer more from work” reading. *Welfare falls*.

(ii) **Higher utility of leisure.** Add an additive bonus $\theta(1 - h)$ to the worker’s payoff. Each unit of leisure gives extra utility on top of the avoided disutility of work. This delivers h^* down AND V_w up. This is the “Europeans enjoy leisure more” reading. *Welfare rises*.

The user’s intuition is (ii), and it is also the right one structurally. “Enjoying life more” should be welfare-improving by construction. We adopt formulation (ii).

This is the standard preference-for-leisure parameter in the cross-country labour-supply literature: Bowles–Park (2005, EJ 115) on emulation and work hours; Olovsson (2009, IER 50(1)) on Europe vs. US; Boppart–Krusell (2020, JPE 128(1)) for the balanced-growth-compatible preference shift framework; Bell–Freeman (2001, JOLE 19(1)) on relative-consumption preferences.

6.2 The model

Worker’s payoff with cultural leisure-premium $\theta^c \geq 0$:

$$V_w(h; \theta^c) = (1 - \tau) w h - \frac{\alpha}{2} h^2 + \theta^c (1 - h), \quad \theta^{EU} > \theta^{US} = 0.$$

The first two terms are the standard quasi-linear payoff. The third term is the cultural leisure bonus: each unit of leisure $1 - h$ delivers θ^c of extra utility. Same α in both countries; firms unchanged.

The worker’s first-best is then:

$$h^*(\tau, \theta^c) = \frac{(1 - \tau) w - \theta^c}{\alpha}, \quad \frac{\partial h^*}{\partial \theta^c} < 0.$$

The worker's payoff at the first-best is $V_w^* = [(1 - \tau)w - \theta^c]^2 / (2\alpha) + \theta^c$, and $dV_w^* / d\theta^c = 1 - h^* > 0$: *higher cultural premium $\Rightarrow$ higher worker welfare at the first-best*, as intuition demands.

6.3 Calibration

We pick $\theta^{US} = 0$ and $\theta^{EU} = 0.156$. The latter is calibrated so that

$$h_{EU}^* = \frac{0.6 - 0.156}{2} = 0.222,$$

matching the empirical EU level used in §2's micro-elasticity decomposition. The 0.156 figure is roughly 26% of the net hourly wage $(1 - \tau)w$ – each EU hour of leisure carries an extra utility worth about a quarter of a market hour, a magnitude consistent with Stevenson–Wolfers (2008).

6.4 Welfare implication: EU is welfare-DOMINANT

Plug in the numbers. At each country's first-best:

$$\text{US: } V_w^* = 0.090, V_f^* = 0.210, W^{US} = 0.300.$$

$$\text{EU: } V_w^* = 0.205, V_f^* = 0.173, W^{EU} = 0.378.$$

The EU worker is strictly better off (V_w rises from 0.090 to 0.205, even though h falls from 0.300 to 0.222): the leisure premium adds direct utility. The EU firm is worse off (lower hours worked). Joint surplus rises ($\Delta W = +0.078$) because the worker's gain dominates the firm's loss. *The cultural premium is a free utility source from the planner's perspective.*

6.5 Why $\theta^{EU} > \theta^{US}$? Three structural candidates

(a) **AGS (2005) historical channel.** Post-WWII union-driven shorter-week reforms in Europe shifted social norms toward a “leisure-is-valuable” equilibrium, and over time those norms intensified leisure preferences (Bisin–Verdier 2001 give the cultural-transmission micro-foundation). Under this reading, *the union channel of §5 is the upstream cause of the cultural channel of §6*: Q5 unions are the proximate coordination device, Q6 leisure preferences are the long-run residue.

(b) **Bowles (1998) endogenous preferences.** Markets shape preferences over generations; long histories of different labour-market institutions produce different aggregate preferences for leisure.

(c) **Stevenson–Wolfers (2008) subjective-wellbeing residual.** Europeans report higher subjective wellbeing per hour of leisure than Americans – the direct empirical anchor for $\theta^{EU} > \theta^{US}$.

6.6 Discriminating evidence

The cleanest test of preference-shift vs. coordination/spillover explanations is whether immigrant workers from low- h countries import home-country preferences (preference shift) or assimilate to host-country institutions (coordination). Empirical surveys (Fernández–Fogli 2009; Algan–Cahuc 2010) find evidence for both, with second-generation immigrants partly retaining home-country preferences but converging over generations. The leisure-bonus specification is consistent with this evidence.

7. Q7 – the welfare comparison

Now we put everything together. Take the EU equilibrium – AGS coordination on h_L plus union bargaining at some $\eta^{EU} > 0$ – and ask: how much of the EU–US welfare gap comes from each of the two channels (culture, union)?

7.1 The four cases

At common τ and common preferences, the EU can sit below the US for one of four reasons, each with a distinct welfare answer:

- (I) US at h^{firm} , EU at h^{soc} via cap or moderate union: *EU dominates*, by exactly the Q4–Q5 mechanism.
- (II) US at h^{firm} , EU below h^* via overshoot cap or excessive union: *EU welfare-decreasing*.
- (III) Same institutions, $\theta^{EU} > \theta^{US}$: each country at its own first-best, but the EU is welfare-dominant because the cultural premium adds free utility from leisure.
- (IV) **Compounded.** Both the institutional bargain and the cultural parameter differ – the empirically realistic case.

7.2 The counterfactual decomposition

We run case (IV) explicitly. Starting from the US baseline ($\theta^{US}=0$, $\eta^{US}=0$ giving $w=1$, $h=0.5$, $W^{US}=0.300$) and shutting down each channel in sequence, we get:

- *Apply the EU cultural shift alone* (raise θ to $\theta^{EU} = 0.156$, keep $\eta = 0$): the firm-power IR-binding wage falls to $w=0.879$ (because the worker accepts lower pay when leisure has direct utility), hours rise to $h=0.561$, but the worker's V_w stays at the IR baseline; firm captures more profit. Total W rises to 0.364, $\Delta W = +0.064$.
- *Add the EU union on top* (raise η to 0.5 at θ^{EU}): the union pulls wages back up to $w=1.172$, hours fall to $h=0.414$. Worker captures more surplus (V_w rises), firm less. Total W rises further to 0.382, $\Delta W = +0.018$.

Net EU–US gap: $\Delta W = +0.082$ – *EU is welfare-dominant*.

The reverse ordering (union first, then culture):

- Apply union alone (raise η to 0.5 at $\theta^{US}=0$): W falls to 0.283, $\Delta W = -0.017$ (canonical RTM result – union extracts wage rents but creates tax-wedge dead-weight).
- Add culture (raise θ to θ^{EU} at $\eta=0.5$): W rises sharply to 0.382, $\Delta W = +0.100$.

Averaging the two orderings (Shapley decomposition) gives the path-independent contributions:

$$\Delta W_{\text{culture}}^{\text{Shapley}} = +0.082, \quad \Delta W_{\text{union}}^{\text{Shapley}} \approx 0.$$

The cultural channel does essentially all of the work. The union channel is welfare-improving in path A (+0.018, because under EU culture the firm-power baseline over-extracts hours and the union pulls them back) and welfare-reducing in path B (−0.017, the canonical RTM result). On average across paths, it nets to roughly zero.

7.3 The reading

This is the substantive answer. The cultural channel does the heavy lifting: when Europeans value leisure more, they enjoy life more, and joint welfare rises. The union channel is small and ambiguous in sign on average. The dominant force in the EU–US labour-market welfare comparison is the cultural premium on leisure, not the bargaining institution.

This contradicts a naive reading of Q5 in isolation (where the canonical RTM model gives unions a welfare-reducing role). Why? Because under the EU cultural premium, firm-power

equilibrium over-extracts hours: the worker would prefer fewer hours at any given wage (higher $\theta \Rightarrow$ lower h^*), so when the firm sets $h = h^d(w)$ at the IR-binding wage, h is too high relative to the worker’s preference. The union pulls h back down toward h^* , which improves welfare. The Q5-only “unions destroy welfare via tax-wedge dead-weight” result still applies in path B (no cultural shift), but path A activates a counter-mechanism.

7.4 The honest welfare-units caveat

Comparing $V_w + V_f$ across countries with different θ^c uses the raw simulation units. The leisure-bonus specification holds α identical and only adds an additive cultural premium, so the V_w comparison is on a common cardinalisation (same monetary numeraire). Mapping to a money-metric (compensating variation in consumption-equivalent units, the Jones–Klenow 2016 standard) leaves the sign and order of magnitude unchanged.

7.5 The bigger picture

Jones–Klenow (2016) report a US–EU welfare gap of -10% to -25% in consumption-equivalent units across all channels (consumption, leisure, life expectancy, inequality). Of these, mortality and aggregate consumption levels carry the bulk – Becker–Philipson–Soares (2005) and Hall–Jones (2007) calibrate a value-of-life such that the US–EU life-expectancy gap of ~ 4 years contributes a welfare *gain* for Europe of roughly $+20\%$. The labour-market block we analyse here is therefore sign-determinate but quantitatively second-order to demographic, TFP, and subjective-wellbeing channels that this analysis does not address. We are answering “how much of the labour-market gap can each institutional channel explain?”, not “is Europe better off than the US overall?”.

8. Where things stand

The take-home is now four blocks (Q1–Q2–Q4–Q5–Q6–Q7) that share a common firm-worker framework and answer one question each:

- Q1 says the data is suggestive but noisy.
- Q2 says the simplest model fits at an implausibly large elasticity, so we need other channels to explain at least half the cross-country gap.
- Q4 explores one such channel (hours regulation) and derives the welfare-improving region for a cap.
- Q5 explores another (a single industry-wide union) and derives the welfare-improving region for the bargaining strength η .
- Q6 explores a third (a leisure bonus that adds direct utility from leisure, $\theta^{EU} > \theta^{US} = 0$, after Bowles–Park 2005 / Olovsson 2009 / Boppart–Krusell 2020 / Stevenson–Wolfers 2008) and shows that “Europeans enjoy life more” is genuinely welfare-improving by construction.
- Q7 combines them and reports the welfare gap of EU vs. US under the canonical model: $\Delta W = +0.082$ (*EU welfare-dominant*), with the cultural channel doing essentially all of the work and the union channel netting to zero in the Shapley average.

Source files: `problem1.tex` (the submission), `research/build_figure_panel.py` (Q1 figure), `research/sim_models_v2.py` (Q2 decomposition, Q5 RTM, Q6 leisure-bonus culture, Q7 layered counterfactual).